

Mia Sumera

Atlanta, GA | www.linkedin.com/in/mia-sumera

Education

Rollins School of Public Health, Emory University Atlanta, GA
Master of Public Health, Epidemiology – Data Science Certificate May 2026

- **Relevant coursework:** *Statistical Programing (EPI 534), Introduction to R (BIOS 544), Applied Machine Learning (DATA 534), Introduction to GIS (DATA 530), Data Science Toolkit (DATA 550), Python Programming (BIOS 584), Current Topics in Data Science (DATA 555), Critical Issues in Global Health (GH 500)*

Clemson University Clemson, SC
Bachelor of Science, Biological Sciences – Minors in Psychology and Microbiology May 2024

- **Relevant coursework:** *Public Health Microbiology (MICR 4000), Medical Terminology & Communication (HLTH 2600)*

Technical Competencies

Programming Languages: *R, SAS, Bash, Python*

Tools & Software: *Docker, Git, ArcGIS, REDCap, Qualtrics, Excel*

Professional Experience

Hope Givers – Connected Content, Inc. Atlanta, GA
Data & Program Analyst (GRA) September 2025 – Present

- Sole data analyst for the Emmy® Award-winning PBS series *Hope Givers*, interpreting data to guide program strategy, support student wellbeing, and inform stakeholders
- Designed and managed dashboards, analyzed program and audience data from 15,000+ students across 20 Georgia counties, and evaluated outcomes to optimize effectiveness

Heart Health United Atlanta, GA
Graduate Research Assistant October 2024 – Present

- Supported data collection and outcome evaluation to assess the impact of the Soccer for Cardiovascular Health (CVH) promotion program
- Generated data-driven visualizations to translate complex health metrics into actionable insights
- Managed SMS outreach initiatives to maximize participant engagement and study retention

Frost Lab – Nell Hodgson Woodruff School of Nursing, Emory University Atlanta, GA
Graduate Research Associate September 2024 – Present

- Contributed to grant writing and report preparation to secure research funding
- Conducted study design and protocol development to assess dementia caregiver preparedness
- Specialized in data management, ensuring accurate collection, storage, and analysis
- Established a nationwide Community Advisory Board framework, coordinating diverse stakeholder expertise to advance the Frost Lab's long-term research goals

U.S. Naval Research Laboratory Washington, D.C.
Physical Science Technician – Molecular Interfaces & Tribology Division February 2020 – January 2023

- Investigated microbiological degradation and deterioration of polymers
- Developed hands-on proficiency in molecular biology, microbiology, and microscopy
- Applied laboratory techniques spanning microorganism cultivation to spectral data collection

Physical Science Technician – Surface Nanoscience & Sensor Technology Division June 2018 – February 2020

- Optimized a clinical diagnostic system, improving sensitivity and signal by 1000-fold
- Acquired foundational knowledge of surface chemistry, immunology, and microfluidics
- Applied laboratory techniques spanning sensing platform prototyping to electron microscopy
- Received the **Alan Berman Research Publication Award**, recognizing the best technical writing in each scientific division

Leadership Experience

Clemson Humanities Prison Initiative

Clemson, SC

President (2023 – 2024) | Vice President (2022 – 2023)

- Co-founded a student-led organization dedicated to advancing educational equity for incarcerated individuals and expanding community-based learning for Clemson students. Partnered with Common Good Atlanta and Georgia State University to provide volunteer opportunities, cultivate student engagement, and support college degree attainment within prison facilities
- Secured \$500+ in Clemson University funding through competitive organization proposal process
- Established a writing lab and literature class at Greenville County Detention Center

Fraternity and Sorority Life Diversity, Equity, and Inclusion Ambassadors

Clemson, SC

President (2022 – 2023) | Secretary (2021 – 2022)

- Directed operations and cross-council communications for Clemson's Greek community of 6,000+ students. Facilitated educational programming, coordinated campus-wide events, and served as the Greek community's central DEI resource
- Doubled membership and secured ambassador representation across all four Greek councils
- Recognized by Clemson News and selected for interview by the Executive Director of Student Communication

Sigma Kappa Sorority – Kappa Pi Chapter

Clemson, SC

Vice President of Diversity, Equity, Inclusion, & Access (2021 – 2022)

- Orchestrated DEI workshops, educational programming, and community reports while fostering collaborative relationships across the broader Greek community
- Pioneered one of Clemson's first cross-council sorority events, raising \$700+ for Alzheimer's awareness and promoting Greek unity
- Advised Sigma Kappa chapters nationally on establishing the VPDEIA role, directly influencing organizational structure and capacity

Volunteer Experience & Other Involvement

Common Good Atlanta – *Tutor & Teaching Assistant (2022 – Present)*

AnMed Health Medical Center – *Emergency Department Volunteer (2023 – 2024)*

Gentiva Health Services – *Hospice Volunteer (2023 – 2024)*

Charleston AFB 628th Medical Group – *Clinical Shadow & Immersion (2022)*

Clemson Veteran & Paralympic Adaptive Sports Program – *Undergraduate Researcher (2021 – 2024)*

National Alzheimer's Buddies – *Member (2021 – 2024)*

Publications & Presentations

- Mroz, E. L., **Sumera, M.**, Clevenger, C., & Hepburn, K. (2026, July) Caring Again: How Grief from Past Caregiving Shapes Burden in New Dementia Care Roles. To be presented at the International Association of Gerontology and Geriatrics World Congress, Amsterdam, The Netherlands.
- Barlow, D.E., Lu, Q., Kong, L., Montgomery, M.T., Zelik, S., **Sumera, M.**, Roman, V.A., Varaljay, V.A., Crookes-Goodson, W., & Biffinger, J.C., "Hydrolysis of polyester coatings by *Papiliotrema laurentii* under nutrient limitation: Weak organic acid toxicity vs. carbon utilization in the microbial – material transformation pathways," manuscript in press, *Applied and Environmental Microbiology*, 2025.
- Barlow, D.E., Lu, Q., Zelik, S., Roman, V.A., Varaljay, V.A., Biffinger, J.C., Montgomery, M.T., **Sumera, M.**, Wagner, D., & Crookes-Goodson, W.J., "Relationships between fungal physiology and biodegradation of polyester-based coatings by *Papiliotrema laurentii* under nutrient-limited and humidified air incubation," presentation at ASM Microbe, June 9 – 13, 2022, Washington D.C.
- Mulvaney, S.P., Kidwell, D.A., Lanese, J.N., Lopez, R., **Sumera, M.E.**, & Wei, E. Catalytic Lateral Flow Immunoassays (cLFIA): Amplified Signal in a self-contained assay format. *Sensing and Bio-Sensing Research*. 2020; 30:1-11. doi; 10.1016/j.sbsr.2020.100390
- Mulvaney, S.P., Kidwell, D.A., Lanese, J.N., Lopez, R., **Sumera, M.E.**, & Wei, E., "Catalytic Enhancement of Lateral Flow Immunoassays, More Signal Amplified the Opportunity," oral presentation at the Molecular Tri-Conference, March 1 – 4, 2020, San Francisco, CA.
- Mulvaney, S.P., Kidwell, D.A., Lanese, J.N., Lopez, R., **Sumera, M.E.**, & Wei, E., "Catalytic Nanoparticles to Enhance the Sensitivity of Lateral Flow Immunoassays," poster presentation at DTRA Chemical and Biological Defense Science & Technology Conference, November 19-22, 2019, Cincinnati, OH.
- Mulvaney, S.P., Kidwell, D.A., Lanese, J.N., Wei, E., Lopez, R., & **Sumera, M.E.**, "Catalytic Nanoparticles as Labels in Lateral Flow Immunoassays," oral presentation at the Next Generation Diagnostics Summit, August 22, 2019, Washington, D.C.
- Mulvaney, S.P., Lanese, J.N., Krute, C.N., **Sumera, M.**, & Fitzgerald, L.A., "Inbios International, Inc. - Active Plague Detect (APD) Rapid Test," NRL Report submitted to the Defense Threat Reduction Agency, December 3, 2018, POC: Dr. Stephen Francesconi.